Observable Classicalization: Admissibility, Projection, and Quantum Axion Detection

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Abstract

Recent work on the quantum observability of axion dark matter argues that intrinsically quantum features of the dark matter field are operationally undetectable due to mode averaging, thermal noise, and extremely weak coupling between dark matter and cavity observables. The resulting conclusion is that axion dark matter may always be treated as an effectively classical stochastic field for experimental purposes. While the mathematical derivations underlying this conclusion are internally rigorous, the interpretation of these results depends critically upon implicit assumptions concerning observability, projection, and the ontology of measurement.

This paper develops a critique of observable classicalization from the perspective of the Relativistic Scalar–Vector Plenum (RSVP), TARTAN, CLIO, and admissibility-based field frameworks. Rather than interpreting Gaussianization and loss of measurable nonclassicality as evidence that the underlying field behaves classically, the present work argues that these results instead reveal the severe projection constraints imposed by detector topology itself. The detector is interpreted not as a transparent observer of an underlying field, but as a constrained admissibility operator that compresses high-dimensional recursive field dynamics into a narrow observable manifold.

Under this interpretation, weakly coupled cavity observables become examples of projection-induced semantic compression. Observable Gaussianization emerges naturally as an entropy-maximizing attractor under repeated admissibility averaging, while deeper recursive coherence structures may persist in topological, relational, or higher-order invariants inaccessible to local cavity observables. The resulting framework distinguishes ontological classicality from observable classicalization and reinterprets decoherence, mode averaging, and weak coupling as geometric properties of observer-field admissibility relations rather than direct evidence concerning the underlying structure of the dark matter plenum.

1 Introduction

The question of whether axion dark matter behaves classically or quantum mechanically has become increasingly important as quantum-enhanced detection technologies approach regimes of extraordinary sensitivity. Recent analyses of cavity haloscope systems have argued that intrinsically quantum effects of axion dark matter are operationally undetectable even if the underlying field occupies highly nonclassical states. The central argument proceeds through two primary mechanisms. First, realistic detectors couple not to microscopic field modes individually, but to effective aggregate modes formed through weighted averaging across large ensembles of momentum states. Second, the extraordinarily weak coupling between dark matter and detector observables suppresses measurable quantum signatures below experimentally accessible thresholds.

The resulting conclusion is operational rather than directly ontological. The detector cavity behaves as though it were interacting with a classical stochastic field, even if the underlying dark matter state possesses highly nonclassical structure. Observable negativity in the Glauber–Sudarshan P-representation, sub-Poissonian fluctuations, squeezing, and related nonclassical signatures become catastrophically suppressed under realistic experimental conditions. Observable distinctions between classical stochastic ensembles and genuinely quantum states effectively disappear.

The present paper does not dispute the mathematical correctness of these derivations. Rather, it argues that the interpretation of these results depends fundamentally upon assumptions concerning the relationship between observability and ontology. The central claim developed here is that the apparent classicalization of observable quantities arises not necessarily because the underlying dark matter field lacks recursive quantum structure, but because the detector itself acts as a severe projection operator over a much larger admissibility manifold.

Within the RSVP framework, observability is never treated as transparent access to an underlying reality. Instead, every measurement process is interpreted as a constrained admissibility projection operating over a recursive field manifold whose full dimensionality remains inaccessible to bounded observers. Under such conditions, observable manifolds may classicalize even while deeper recursive structures remain highly coherent, nonlocal, and nonclassical.

This distinction between ontological structure and observable projection lies at the center of the critique developed in this work.

The paper proceeds by reinterpreting the cavity haloscope formalism through the lens of admissibility dynamics, recursive field geometry, semantic projection theory, and entropy-constrained observational manifolds. The detector cavity is modeled as a local admissibility kernel that performs geometric compression over incoming field trajectories. Gaussianization then emerges not as evidence that the field itself is fundamentally classical, but as a generic property of entropy-maximizing projection dynamics under severe observational constraints.

This reinterpretation yields several conceptual consequences.

First, observable classicalization becomes a quotient-space phenomenon induced by detector geometry rather than a direct statement concerning the microscopic ontology of dark matter itself.

Second, the quantum central limit behavior derived in existing analyses is reinterpreted as recursive admissibility averaging over trajectory bundles whose higher-order coherence structure may remain preserved beyond the observable manifold.

Third, decoherence is reformulated geometrically as redistribution of accessibility across field topology rather than literal destruction of underlying recursive structure.

Fourth, weak coupling is distinguished from weak global influence. Small local interaction efficiencies may nevertheless generate large-scale admissibility restructuring over sufficiently long cosmological intervals.

Finally, the observer-system separation assumed in standard measurement theory is replaced by coupled recursive observer-field dynamics in which detector admissibility structure coevolves with the projected field manifold itself.

Under this interpretation, the operational success of classical stochastic descriptions does not imply that the underlying field is genuinely classical. Instead, classical observability emerges as a stability attractor of constrained projection dynamics.

The resulting framework therefore reframes the debate surrounding quantum axion observability. The inability of current detectors to reconstruct nonclassical structure does not establish the absence of such structure. Rather, it reveals the geometric limitations of presently realizable projection architectures.

2 Observable Classicalization and Projection Geometry

The conventional interpretation of cavity haloscope measurements implicitly assumes that detector observables provide direct access to the physically relevant structure of the dark matter field. Under this assumption, if all experimentally accessible observables admit an effectively classical stochastic description, then the underlying field may itself be regarded as operationally classical.

The RSVP framework rejects this inference.

Instead, detector observables are interpreted as compressed projections from a much larger recursive field manifold into a constrained admissibility subspace determined by detector topology, interaction geometry, and entropy constraints. Observable quantities therefore represent only a thin projection of the total dynamical structure of the underlying field.

Let

$$X$$

denote the full recursive configuration manifold of the dark matter plenum. Let

$$M$$

denote the observable manifold accessible to a detector cavity. The measurement process is represented by a projection operator

$$\pi : X \to M.$$

The observables extracted experimentally are therefore not direct observables on the underlying field manifold itself, but projected observables:

$$\mathcal{O}_{\text{obs}} = \pi_*(\mathcal{O}_X).$$

This distinction is fundamental.

The cavity formalism demonstrates that

$$\pi_*(\rho_X)$$

admits Gaussian stochastic descriptions under mode averaging and weak coupling. However, this does not imply that

$$\rho_X$$

itself possesses Gaussian or classical structure. Multiple inequivalent recursive quantum configurations may project into identical observable statistics.

The detector therefore induces an admissibility quotient structure over the underlying field manifold. Entire sectors of recursive phase information become observationally annihilated under projection.

From this perspective, the observed emergence of classical stochastic behavior is not surprising. Gaussianization is a generic consequence of entropy-maximizing projection under coarse-grained admissibility compression.

The detector does not reveal the underlying field directly.

It constructs a locally admissible approximation to the field consistent with the detector's finite resolution geometry.

3 Recursive Admissibility and Detector Compression

The cavity formalism used in contemporary axion analyses may be reinterpreted naturally within the RSVP framework as an admissibility compression process operating over recursive field trajectories.

Let the dark matter plenum be described by the coupled RSVP field triple

$$(\Phi, \mathbf{v}, S),$$

where:

$$\Phi$$

represents scalar field density,

$$\mathbf{v}$$

represents directed field flow, and

$$S$$

represents configurational accessibility entropy.

The local admissibility of a field configuration is defined by

$$\mathcal{A}(x,t) = e^{-S(x,t)}.$$

Under this interpretation, coherent field organization corresponds not merely to amplitude structure, but to persistence of low-entropy admissible trajectories through recursive field evolution.

The detector cavity imposes a local admissibility kernel

$$K_D(x,p)$$

over incoming field trajectories. The observed effective mode therefore becomes

$$a_{\text{eff}} = \int K_D(x,p)\Psi(x,p)\,dx\,dp,$$

where

$$\Psi(x,p)$$

represents the underlying recursive field configuration over position and momentum space.

This expression parallels the effective mode averaging used in standard cavity analyses, but its interpretation differs substantially.

Conventional treatments regard effective mode averaging as evidence that microscopic quantum structure becomes operationally irrelevant. RSVP instead interprets the averaging process geometrically. The detector performs entropy-constrained smoothing over the incoming admissibility manifold.

Observable Gaussianization therefore arises because the detector preferentially stabilizes entropy-maximizing admissible approximations within its finite observable geometry.

The detector acts as a semantic compression operator.

This reinterpretation changes the meaning of classicalization itself. The cavity does not establish that the underlying field lacks recursive coherence. Instead, the detector selectively annihilates large sectors of admissibility structure whose dimensionality exceeds the detectorâĂŹs observable manifold.

The effective entropy of the projected observable state therefore satisfies

$$S_{\text{obs}} \geq S_{\text{micro}},$$

with equality only when the detector resolves the full recursive trajectory bundle of the incoming field.

In practice, realistic cavity detectors possess extremely low-dimensional observational geometry relative to the complexity of the underlying field manifold. Observable classicalization therefore emerges naturally as a consequence of severe admissibility compression.

4 The Quantum Central Limit Theorem as Geometric Averaging

One of the most important claims of recent axion analyses is that the detectorâĂŹs effective mode becomes approximately Gaussian through aggregation across many independent microscopic dark matter modes. The resulting application of the quantum central limit theorem plays a central role in the argument for operational classicality.

Within RSVP, this result is not disputed mathematically. However, the interpretation of the theorem changes substantially.

Suppose the underlying microscopic field modes are represented by

$$\{\psi_i\}_{i=1}^{N}.$$

The detector forms the effective observable mode

$$\Psi_{\text{eff}} = \sum_i w_i \psi_i,$$

where

$$w_i$$

are detector-dependent coupling weights determined by cavity geometry.

Under sufficiently weak intermode correlations, the projected observable manifold approaches Gaussian statistics:

$$\lim_{N\to\infty} \pi_*(\Psi_{\text{eff}}) = \mathcal{N}(0, \sigma^2).$$

Conventional interpretations often treat this result as evidence that the underlying field effectively behaves classically. RSVP instead treats it as a projection theorem concerning observable manifolds only.

The critical distinction is that low-order observables need not preserve higher-order recursive coherence structures.

Define the recursive coherence tensor

$$\mathcal{C}_{ijkl} = \langle \psi_i \psi_j \psi_k \psi_l \rangle.$$

Standard cavity observables depend primarily upon low-order moments such as

$$\langle \psi_i \rangle$$

and

$$\langle \psi_i \psi_j \rangle.$$

However, recursive field structure may persist in higher-order invariants inaccessible to local cavity projections.

Define generalized recursive invariants

$$\mathcal{I}_n = \mathrm{Tr}(\mathcal{C}^n).$$

Observable Gaussianization may suppress low-order deviations from classicality while leaving these higher-order recursive structures largely intact.

The detector therefore induces observational flattening over the field manifold.

Under repeated admissibility averaging, many inequivalent recursive topologies collapse into identical Gaussian observable statistics.

Observable classicalization becomes a property of the projection architecture rather than of the underlying field ontology.

5 Topological Persistence and Hidden Coherence

The standard cavity formalism treats coherence primarily through local phase relationships and low-order statistical observables. RSVP instead interprets coherence geometrically through persistence structure over recursive trajectory bundles.

Let the dark matter field occupy a recursive bundle

$$\mathcal{B} = (X, \mathcal{T}, \omega),$$

where:

$$X$$

is the field manifold,

$$\mathcal{T}$$

is trajectory topology, and

$$\omega$$

represents recursive phase organization.

Observable measurements induce a projection

$$\pi : \mathcal{B} \to \mathcal{B}_{\mathrm{obs}}.$$

The critical issue is whether this projection preserves the persistence structure of the original bundle.

Define the persistence functional

$$P_k(\epsilon) = \operatorname{rank} H_k(\mathcal{B}_\epsilon),$$

where

$$H_k$$

denotes the k-th homology group over an admissibility filtration parameterized by

$$\epsilon.$$

Under severe admissibility compression,

$$P_k^{\text{obs}}(\epsilon) \ll P_k(\epsilon).$$

The detector therefore destroys access to large sectors of topological coherence even if the underlying recursive field structure remains intact.

This provides an alternative interpretation of decoherence.

Conventional formulations treat decoherence as destruction of quantum structure through environmental interaction. RSVP instead interprets decoherence geometrically as redistribution of accessibility across the field manifold.

Local observables lose coherence not necessarily because coherence ceases to exist, but because coherence migrates into inaccessible regions of recursive trajectory topology.

Observable decoherence is therefore projection-relative.

The cavity detector measures only those sectors of recursive organization compatible with its local admissibility geometry.

6 Weak Coupling and Recursive Influence

One of the strongest arguments advanced in contemporary axion detection theory concerns the extraordinarily weak coupling between dark matter and detector observables. Observable quantum signatures become suppressed by powers of the interaction efficiency

$$\eta,$$

which for realistic cavity haloscopes may become vanishingly small.

From the perspective of standard measurement theory, this suppression appears decisive. If quantum signatures scale proportionally to

$$\eta,$$

then experimentally accessible observables become operationally classical.

The RSVP framework introduces a crucial distinction between local interaction strength and global admissibility influence.

Weak coupling does not necessarily imply weak structural effect.

Instead, RSVP interprets interaction as recursive admissibility modulation unfolding over extended temporal and geometric scales.

Define the local interaction efficiency

$$\eta(x,t).$$

Conventional interpretations effectively assume:

$$\eta \to 0 \quad \Rightarrow \quad \text{physical irrelevance.}$$

RSVP rejects this implication.

Introduce the recursive influence functional

$$\mathcal{R}(t) = \int_0^t \eta(\tau)\Gamma(\tau)\,d\tau,$$

where

$$\Gamma(\tau)$$

measures admissibility leverage density over recursive trajectory evolution.

Even if

$$\eta \ll 1,$$

the cumulative restructuring of admissibility topology may become substantial over sufficiently large temporal intervals:

$$\lim_{t\to\infty} \mathcal{R}(t) \gg 1.$$

This distinction mirrors broader RSVP principles concerning constraint dynamics. Small energetic perturbations may nevertheless reorganize global trajectory structure if they act coherently over large recursive manifolds.

Weak local observability therefore does not imply weak cosmological influence.

Indeed, RSVP predicts that many large-scale structural phenomena emerge not through strong local forces, but through recursive admissibility steering operating across extremely long dynamical horizons.

The cavity formalism therefore establishes only that local detector observables weakly sample the incoming field. It does not establish that the field itself possesses weak global organizational influence.

7 Detector Geometry as an Entropy-Minimizing Operator

The detector cavity may be modeled more generally as an entropy-constrained admissibility optimization system.

Suppose the underlying field configuration is

$$\Psi.$$

The detector does not directly reconstruct

$$\Psi.$$

Instead, it selects the lowest-entropy admissible approximation compatible with its finite measurement geometry.

Define the detector response functional

$$\mathcal{D}[\Psi] = \arg\min_{\phi \in M} \left(\|\Psi - \phi\|^2 + \lambda S(\phi) \right),$$

where:

$$M$$

is the observable manifold,

$$S(\phi)$$

is configurational accessibility entropy, and

$$\lambda$$

controls admissibility weighting.

Observable classicalization then becomes a variational consequence of detector optimization.

Gaussian effective modes emerge because they minimize free-energy-like admissibility functionals under repeated coarse-grained projection.

Define the generalized detector free energy

$$\mathcal{F}[\phi] = E[\phi] + \lambda S[\phi].$$

The detector therefore converges toward:

$$\phi^* = \arg\min_{\phi} \mathcal{F}[\phi].$$

Gaussian stochastic observables become stable attractors because they maximize entropy while remaining compatible with weakly constrained cavity observables.

This interpretation transforms the meaning of classicality itself.

Observable classicality is not necessarily a fundamental property of the incoming field.

Rather, it is the stable fixed point of entropy-constrained admissibility projection.

The detector constructs the most statistically stable low-dimensional approximation available within its finite observational geometry.

8 Recursive Observer-Field Dynamics

Standard cavity analyses implicitly maintain a separation between observer and observed system. The detector is treated as an external apparatus extracting information from a preexisting dark matter field.

RSVP rejects this separation.

Observer and field instead participate in coupled recursive admissibility dynamics.

Let

$$\mathcal{U}_D$$

denote detector state space and

$$\mathcal{U}_{DM}$$

denote the dark matter field manifold.

Measurement induces coupled evolution:

$$\frac{d}{dt}\begin{pmatrix}\mathcal{U}_D \\ \mathcal{U}_{DM}\end{pmatrix} = \mathcal{L}\begin{pmatrix}\mathcal{U}_D \\ \mathcal{U}_{DM}\end{pmatrix},$$

where:

$$\mathcal{L} = \mathcal{L}_D + \mathcal{L}_{DM} + \mathcal{L}_{\text{int}}.$$

The observable state therefore depends upon recursive interaction history.

Measurement outcomes become path dependent:

$$\mathcal{O}(t) = \mathcal{P}\exp\left(\int_0^t \mathcal{L}_{\text{int}}\, d\tau\right),$$

where

$$\mathcal{P}$$

denotes path ordering.

This interpretation has several consequences.

First, the detector cannot be regarded as a passive classical observer.

Second, observable quantities emerge jointly from detector-field coevolution rather than from unilateral extraction of preexisting information.

Third, observable classicalization may reflect stabilization properties of the combined observer-field system rather than intrinsic properties of the dark matter field alone.

The detector and field recursively constrain one another.

Observable manifolds therefore arise dynamically through admissibility coevolution.

9 The Limits of Operationalism

The cavity formalism ultimately adopts an operationalist stance. If no realistic experiment can distinguish a nonclassical field from a classical stochastic ensemble, then the classical description is treated as sufficient.

RSVP distinguishes sharply between operational sufficiency and ontological equivalence.

The inability to reconstruct nonclassical structure from a constrained observable manifold does not establish that such structure is absent.

Instead, it establishes only that the projection operator associated with the detector annihilates the relevant invariants before measurement occurs.

This distinction may be expressed schematically:

$$\text{observable classicality} \neq \text{ontological classicality}.$$

The detector cavity accesses only a low-dimensional projection of the full recursive field manifold.

Many inequivalent recursive configurations therefore remain observationally degenerate.

The apparent success of classical stochastic descriptions emerges because the detector compresses the incoming field into entropy-maximizing admissible observables whose geometry strongly suppresses recoverable nonclassical structure.

The resulting operational indistinguishability should therefore not be interpreted as evidence that the deeper field ontology is fundamentally classical.

Rather, it reveals the severe geometric limitations of presently realizable projection architectures.

10 Conclusion

The contemporary cavity-haloscope analysis of axion dark matter establishes an important and mathematically rigorous result. Weak coupling, mode averaging, thermal noise, and detector geometry strongly suppress observable signatures of nonclassicality within presently realizable experiments.

The RSVP critique developed here does not dispute these derivations.

Instead, it challenges the ontological interpretation often attached to them.

The central argument of this paper is that observable classicalization is fundamentally a projection phenomenon. Detector cavities operate as entropy-constrained admissibility kernels over a vastly larger recursive field manifold whose full structure remains inaccessible to bounded observers.

Under repeated admissibility averaging, Gaussian stochastic observables emerge naturally as stable low-dimensional attractors.

The resulting effective classicality therefore reflects the geometry of the detector projection rather than necessarily the ontology of the underlying field itself.

From the RSVP perspective, the cavity formalism demonstrates the classicalization of observable manifolds under severe projection constraints. It does not demonstrate that the dark matter plenum itself lacks recursive quantum structure, higher-order coherence, or topological persistence inaccessible to local observables.

This distinction between observability and ontology is essential.

Weakly coupled recursive field systems may possess enormous hidden structure while remaining operationally classical under constrained observational geometries.

The resulting picture reframes the relationship between measurement, decoherence, and classicality itself. Observable reality becomes not a transparent reflection of underlying structure, but a recursively stabilized admissibility projection arising from the geometric limitations of bounded observers embedded within a far larger dynamical plenum.

Appendices

A Projection Operators and Observable Classicalization

The central claim of the critique is that the effective classicality derived in the cavity-haloscope framework arises from projection-induced admissibility collapse rather than from the absence of underlying nonclassical structure. We therefore formalize the detector as a projection operator acting on a higher-dimensional field manifold.

Let

$$X$$

denote the full recursive field configuration space of the dark matter plenum, and let

$$M$$

denote the detector-accessible observable manifold. Define the detector projection

$$\pi : X \to M.$$

The measurable cavity observables are therefore not direct observables on X, but pullbacks along π:

$$\mathcal{O}_{\mathrm{obs}} = \pi_* \mathcal{O}_X.$$

The paper effectively proves that

$$\pi_*(\rho_X)$$

admits a Gaussian stochastic representation under weak coupling and mode averaging. However, this does not imply that

$$\rho_X$$

itself is Gaussian or classical.

Define the admissibility degeneracy class

$$[\rho_X]_\pi = \{\rho'_X \mid \pi_*(\rho'_X) = \pi_*(\rho_X)\}.$$

Then many inequivalent recursive quantum states may map to identical detector statistics. Observable classicalization is therefore a quotient-space phenomenon:

$$X/\ker(\pi_*).$$

The detector effectively annihilates large sectors of relational phase information.

B Recursive Admissibility and Entropic Compression

RSVP interprets measurement not as passive observation but as entropy-constrained admissibility reduction.

Let the dark matter field be represented by the coupled RSVP fields

$$(\Phi, \mathbf{v}, S),$$

where:

$$\Phi$$

is scalar field density,

$$\mathbf{v}$$

is vector flow, and

$$S$$

is configurational accessibility entropy.

Define admissibility as:

$$\mathcal{A}(x,t) = \exp(-S(x,t)).$$

The detector cavity imposes a local admissibility kernel

$$K_D(x,p)$$

over incoming field trajectories.

The measured effective mode is then:

$$a_{\text{eff}} = \int K_D(x,p)\,\Psi(x,p)\,dx\,dp.$$

This reproduces the paper's effective-mode averaging structure, but RSVP interprets the resulting Gaussianization as an entropic contraction:

$$\Psi \mapsto K_D * \Psi.$$

The detector therefore acts as a semantic smoothing operator over recursive field trajectories.

The resulting effective entropy satisfies:

$$S_{\text{obs}} \geq S_{\text{micro}},$$

with equality only if the detector resolves the full admissible trajectory bundle.

C Generalized Quantum Central Limit Collapse

The paper relies heavily on a quantum central limit theorem argument. RSVP reframes this geometrically.

Suppose the microscopic field modes are:

$$\{\psi_i\}_{i=1}^N.$$

The detector forms the weighted aggregate:

$$\Psi_{\text{eff}} = \sum_i w_i \psi_i.$$

If the detector topology cannot preserve long-range recursive correlations, then higher-order phase structure is suppressed under repeated admissibility averaging.

Define the recursive coherence tensor:

$$\mathcal{C}_{ijkl} = \langle \psi_i \psi_j \psi_k \psi_l \rangle.$$

Standard cavity observables depend primarily on low-order moments:

$$\langle \psi_i \rangle, \quad \langle \psi_i \psi_j \rangle.$$

But RSVP proposes that nonclassical structure may persist in higher-order recursive invariants:

$$\mathcal{I}_n = \text{Tr}(\mathcal{C}^n).$$

The central limit theorem suppresses low-order observable deviations while leaving high-order recursive topology intact.

Thus:

$$\lim_{N\to\infty} \pi_*(\Psi_{\text{eff}}) = \mathcal{N}(0, \sigma^2),$$

does not imply:

$$\Psi_X \sim \mathcal{N}(0, \sigma^2).$$

Only the projected manifold classicalizes.

D Topological Persistence of Nonclassical Structure

Let the dark matter field occupy a recursive trajectory bundle:

$$\mathcal{B} = (X, \mathcal{T}, \omega),$$

where:

$$\mathcal{T}$$

is trajectory topology and

$$\omega$$

is recursive phase structure.

Define a persistence functional:

$$P_k(\epsilon) = \operatorname{rank} H_k(\mathcal{B}_\epsilon),$$

where

$$H_k$$

is the k-th homology group of the admissibility filtration.

Then observable classicalization corresponds not necessarily to destruction of topological coherence, but to failure of detector observables to preserve the filtration.

The detector induces:

$$\pi : \mathcal{B} \to \mathcal{B}_{\text{obs}},$$

with:

$$P_k^{\text{obs}}(\epsilon) \ll P_k(\epsilon).$$

The observed field therefore appears classical because the detector collapses persistence structure below the resolution threshold.

E Weak Coupling and Recursive Influence

The paper treats weak coupling as evidence that intrinsically quantum signatures become unobservable.

RSVP distinguishes energetic coupling from admissibility leverage.

Let:

$$\eta$$

be the local interaction efficiency.

Standard analysis assumes:

$$\eta \to 0 \quad \Rightarrow \quad \text{physical irrelevance.}$$

However RSVP proposes recursive accumulation dynamics:

$$\mathcal{R}(t) = \int_0^t \eta(\tau)\,\Gamma(\tau)\,d\tau,$$

where:

$$\Gamma$$

measures trajectory influence density.

Small local interactions may therefore accumulate into large-scale admissibility restructuring over cosmological intervals:

$$\lim_{t \to \infty} \mathcal{R}(t) \gg 1$$

even when

$$\eta \ll 1.$$

Weak coupling suppresses instantaneous observability without necessarily suppressing global geometric influence.

F Detector Geometry as an Admissibility Kernel

The cavity detector may be modeled as a constrained admissibility operator.

Define the detector response functional:

$$\mathcal{D}[\Psi] = \arg\min_{\phi \in M} \left(\|\Psi - \phi\|^2 + \lambda S(\phi) \right).$$

The detector therefore selects the lowest-entropy admissible approximation inside its observable manifold.

Observable classicality becomes the solution of a constrained variational problem:

$$\phi^* = \arg\min_{\phi} \mathcal{F}[\phi],$$

where:

$$\mathcal{F} = E + \lambda S.$$

Gaussian effective modes arise because they are entropy-maximizing stable attractors under repeated admissibility projection.

G Recursive Observer-Field Coupling

The paper assumes observer and field remain separable.

RSVP instead treats measurement as coupled recursive dynamics.

Let:

$$\mathcal{U}_D$$

be detector state space and

$$\mathcal{U}_{DM}$$

the dark matter field manifold.

Measurement induces coupled evolution:

$$\frac{d}{dt} \begin{pmatrix} \mathcal{U}_D \\ \mathcal{U}_{DM} \end{pmatrix} = \mathcal{L} \begin{pmatrix} \mathcal{U}_D \\ \mathcal{U}_{DM} \end{pmatrix},$$

where:

$$\mathcal{L} = \mathcal{L}_D + \mathcal{L}_{DM} + \mathcal{L}_{\text{int}}.$$

The observable state is therefore path dependent:

$$\mathcal{O}(t) = \mathcal{P} \exp \left(\int_0^t \mathcal{L}_{\text{int}} \, d\tau \right).$$

The detector cannot be treated as an external classical observer because its admissibility structure coevolves with the projected field manifold.

H Conclusion of the Mathematical Critique

The axion paper rigorously demonstrates that presently realizable cavity observables classicalize under weak coupling and coarse-grained projection. RSVP does not dispute this derivation.

However, the theory challenges the implicit ontological conclusion drawn from it.

The mathematical structure developed here suggests instead that:

$$\text{observable classicality} \neq \text{ontological classicality}.$$

The detector projects a high-dimensional recursive field manifold into a constrained admissibility subspace whose geometry suppresses measurable nonclassical invariants.

The resulting Gaussianization is therefore interpreted not as destruction of quantum structure, but as projection-induced semantic compression arising from detector topology itself.

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